FLASQUE MODULES ARE ACYCLIC FOR THE GLOBAL SECTION FUNCTOR AND FOR SECTIONS OVER ANY OPEN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written: the maximality argument, the correction using flasqueness of the subsheaf and the final gluing are all present in the source.

Editorial note. The counted spans carry 2 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-flasque-acyclic is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Criterion for a class of objects to be right acyclic. Derived Categories, Lemma derived-lemma-subcategory-right-acyclics, invoked at proof lines 1920-1921. Supplies that the two verified conditions on the class of flasque modules imply acyclicity.
- Every O_X-module embeds into a flasque module. Injectives, Lemma injectives-lemma-abelian-sheaves-space, invoked at proof lines 1922-1924. Supplies the embedding condition, since injective modules are flasque.
- The sheaf condition on G for a union of compatible sections. Standard sheaf theory, used at proof lines 1941-1943 and 1953-1955. Supplies the existence of the glued section over a union of opens.

Source reference. Original lemma 05T8: sources/stacks-chow/tags.txt, source lines 7472–7472. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01DG: sources/stacks-chow/tags.txt, source lines 1797–1797. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Any flasque $\mathcal{O}_X$-module is acyclic for $R\Gamma(X, -)$ as well as $R\Gamma(U, -)$ for any open U of X.*

Proof. We will prove this using Derived Categories, Lemma 05T8. Since every injective module is flasque we see that we can embed every $\mathcal{O}_X$-module into a

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flasque module, see Injectives, Lemma 01DG. Thus it suffices to show that given a short exact sequence

$$0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$$

with $\mathcal{F}$, $\mathcal{G}$ flasque, then $\mathcal{H}$ is flasque and the sequence remains short exact after taking sections on any open of X. In fact, the second statement implies the first. Thus, let $U \subset X$ be an open subspace. Let $s \in \mathcal{H}(U)$. We will show that we can lift s to a section of $\mathcal{G}$ over U. To do this consider the set T of pairs (V, t) where $V \subset U$ is open and $t \in \mathcal{G}(V)$ is a section mapping to $s|_V$ in $\mathcal{H}$. We put a partial ordering on T by setting $(V, t) \leq (V', t')$ if and only if $V \subset V'$ and $t'|_V = t$. If (V_α, t_α), $\alpha \in A$ is a totally ordered subset of T, then $V = \bigcup V_\alpha$ is open and there is a unique section $t \in \mathcal{G}(V)$ restricting to t_α over V_α by the sheaf condition on $\mathcal{G}$. Thus by Zorn's lemma there exists a maximal element (V, t) in T. We will show that $V = U$ thereby finishing the proof. Namely, pick any $x \in U$. We can find a small open neighbourhood $W \subset U$ of x and $t' \in \mathcal{G}(W)$ mapping to $s|_W$ in $\mathcal{H}$. Then $t'|_{W \cap V} - t|_{W \cap V}$ maps to zero in $\mathcal{H}$, hence comes from some section $r' \in \mathcal{F}(W \cap V)$. Using that $\mathcal{F}$ is flasque we find a section $r \in \mathcal{F}(W)$ restricting to r' over $W \cap V$. Modifying t' by the image of r we may assume that t and t' restrict to the same section over $W \cap V$. By the sheaf condition of $\mathcal{G}$ we can find a section $\tilde{t}$ of $\mathcal{G}$ over $W \cup V$ restricting to t and t'. By maximality of (V, t) we see that $V \cup W = V$. Thus $x \in V$ and we are done. $\square$